

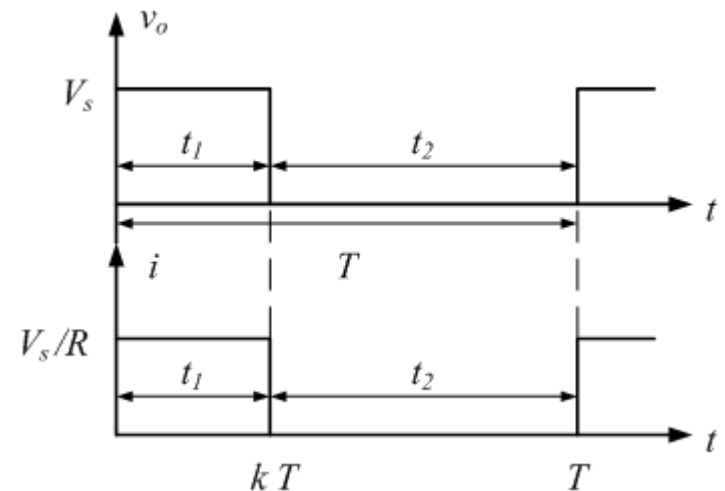
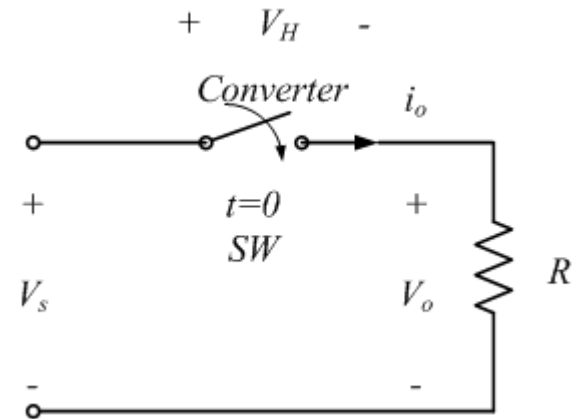
Power Electronics and Motor Drives Lecture 5

DC – DC converters

- They are known as DC-converters and they convert fixed DC directly to variable DC.
- DC converters $\equiv$ AC transformers with variable turns ratio.

Principle of step-down operation

- When the switch SW, Known as the chopper, is closed for a time t_1 , the input voltage V_s appears across the load.
- If the switch remains off for time t_2 , the voltage across the load is zero.
- The switch may be:
 - BJT
 - MOSFET
 - GTO
 - IGBT
- The practical devices have a finite voltage drop from 0.5 to $2V$.
- For the sake of simplicity, the voltage drops across power semiconductor devices are neglected.



- The average output voltage is given by:

$$V_a = \frac{1}{T} \int_0^{t_1} v_o dt = \frac{t_1}{T} V_s = f t_1 V_s = k V_s$$

- The average Load current is given by:

$$I_a = \frac{V_a}{R} = \frac{k V_s}{R}$$

Where, T is the chopping period; $k=t_1/T$ is the duty cycle of chopper; and f is the chopping frequency.

- The rms value of output voltage is found from:

$$V_{o_rms} = \left(\frac{1}{T} \int_0^{kT} v_o^2 dt \right)^{1/2} = \sqrt{k} V_s$$

- Assuming a lossless converter, the input power to the converter is the same as the output power and is given by:

$$P_i = \frac{1}{T} \int_0^{kT} v_o i dt = \frac{1}{T} \int_0^{kT} \frac{v_o^2}{R} dt = k \frac{V_s^2}{R} = \frac{V_{o_rms}^2}{R}$$

- The effective input resistance seen by the source is :

$$R_i = \frac{V_s}{I_s} = \frac{V_s}{I_a} = \frac{V_s}{kV_s/R} = \frac{R}{k}$$

- The duty cycle $0 < k < 1 \rightarrow$ the output voltage $0 < V_o < V_s$.

- There are two type of switching operation:

1- Constant-frequency operation:

The chopping frequency f (or chopping period T) is kept constant and the on-time t_1 is varied. The width of the pulse is varied and this type of control is known as pulse-width-modulation (PWM) control.

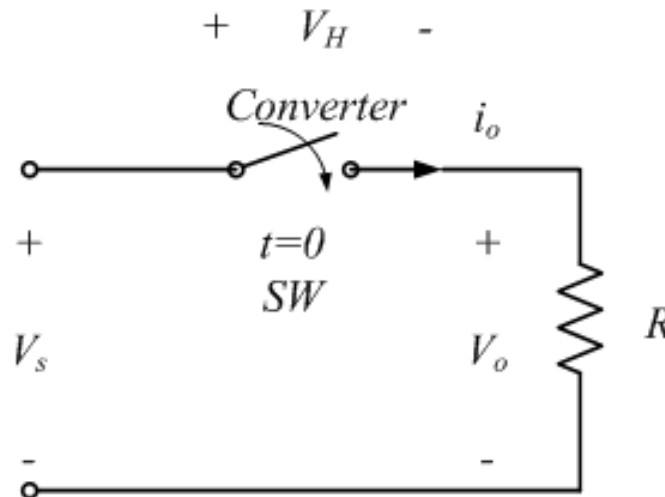
2- Variable-frequency operation:

The chopping frequency f is varied. Either on-time t_1 or off-time t_2 is kept constant. This is called frequency modulation. The frequency has to be varied over a wide range to obtain the full output voltage range. This type of control would generate harmonics at unpredictable frequencies and the filter design would be difficult.

Example 1:

The dc converter shown in the figure below has a resistive load of $R=10\Omega$ and the input voltage is $V_s=220V$. When the converter switch remains on, its voltage drop is $v_{ch}=2V$ and the chopping frequency is $f=1kHz$. If the duty cycle is 50% , determine

- (a) the average output voltage V_a ,
- (b) the rms output voltage V_o ,
- (c) the converter efficiency,
- (d) the effective input resistance R_i of the converter.



Solution:

$$R=10\ \Omega, V_s=220V, k=0.5, v_{ch}=2V, f=1kHz.$$

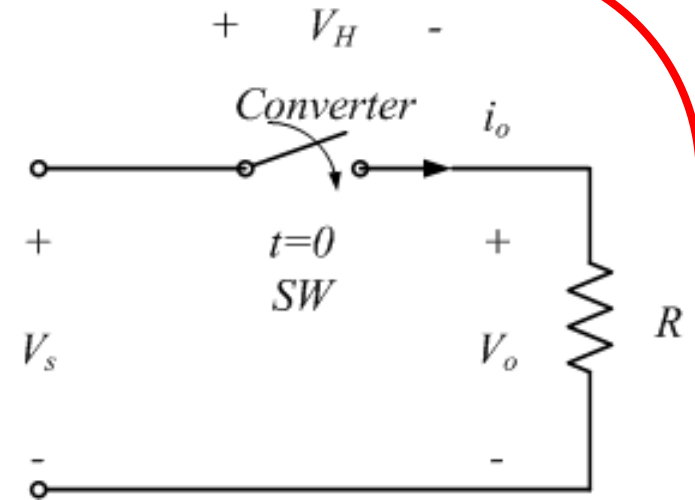
$$(a) V_a = kV_s = 0.5 \times (220 - 2) = 109V.$$

$$(b) V_o = \sqrt{k}V_s = \sqrt{0.5} \times (220 - 2) = 154.15V$$

(c) The output power can be found from:

$$P_o = \frac{1}{T} \int_0^{kT} \frac{V_o^2}{R} dt = \frac{1}{T} \int_0^{kT} \frac{(V_s - V_{ch})^2}{R} dt = k \frac{(V_s - V_{ch})^2}{R}$$

$$P_o = 0.5 \times \frac{(220 - 2)^2}{10} = 2376.2W$$



The input power to the converter can be found from

$$P_i = \frac{1}{T} \int_0^{kT} V_s i dt = \frac{1}{T} \int_0^{kT} \frac{V_s (V_s - V_{ch})}{R} dt = k \frac{V_s (V_s - V_{ch})}{R}$$

$$P_i = 0.5 \times 220 \times \frac{220 - 2}{10} = 2398 \text{ W}$$

The converter efficiency is

$$\eta = \frac{P_o}{P_i} = \frac{2376.2}{2398} = 99.09\% = \frac{V_s - V_{ch}}{V_s}$$

$$(d) R_i = \frac{V_s}{I_a} = \frac{V_s}{k(V_s - V_{ch})/R} = \frac{R}{k(1 - V_{ch}/V_s)} = \frac{10}{0.5(1 - 2/220)} = 20.18 \Omega$$

Generation of duty cycle

➤ The duty cycle k can be generated by comparing a dc reference V_r with a sawtooth carrier signal V_{cr} .

➤ The carrier signal V_{cr} is given by:

$$V_{cr} = \frac{V_{rp}}{T} t$$

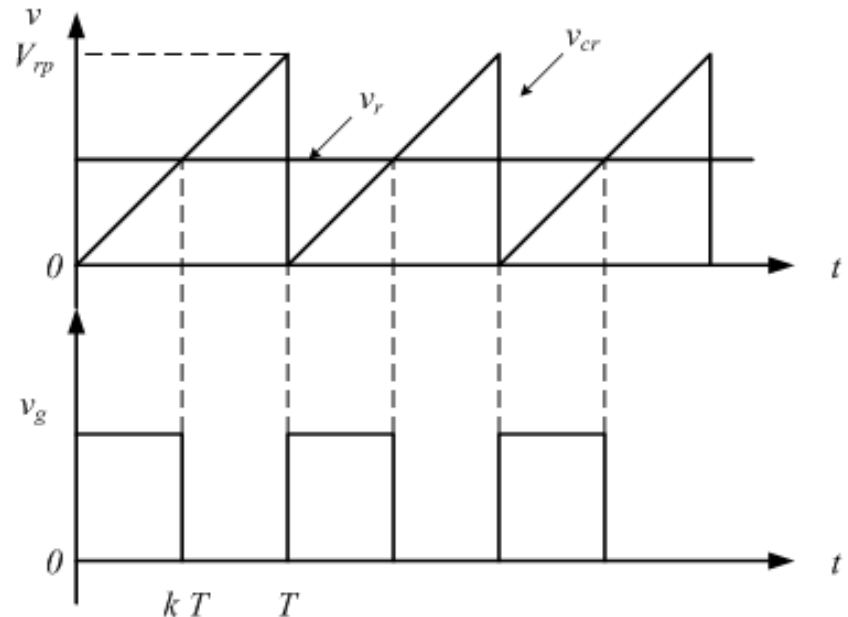
At $t = kT$,

$$V_r = V_{cr}$$

$$V_r = \frac{V_{rp}}{T} kT$$

$$k = \frac{V_r}{V_{rp}}$$

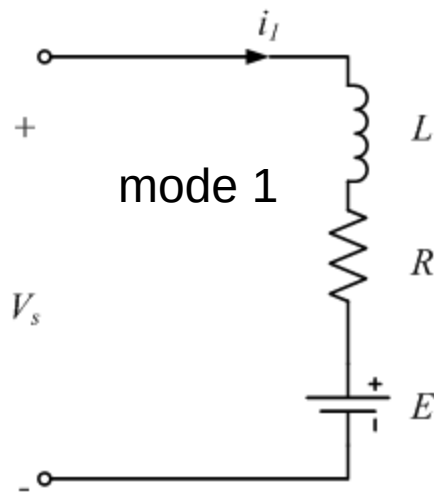
➤ By varying V_r from 0 to V_{rp} , the duty cycle k can be varied from 0 to 1.



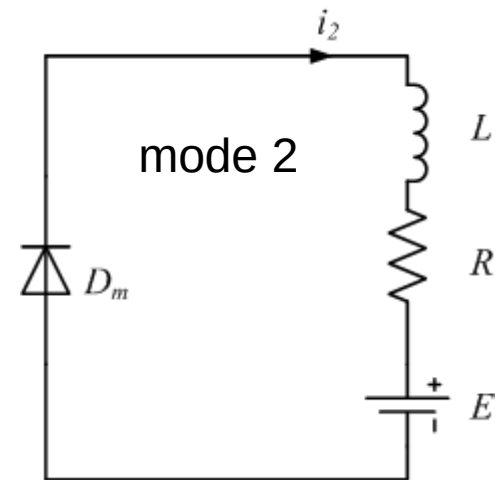
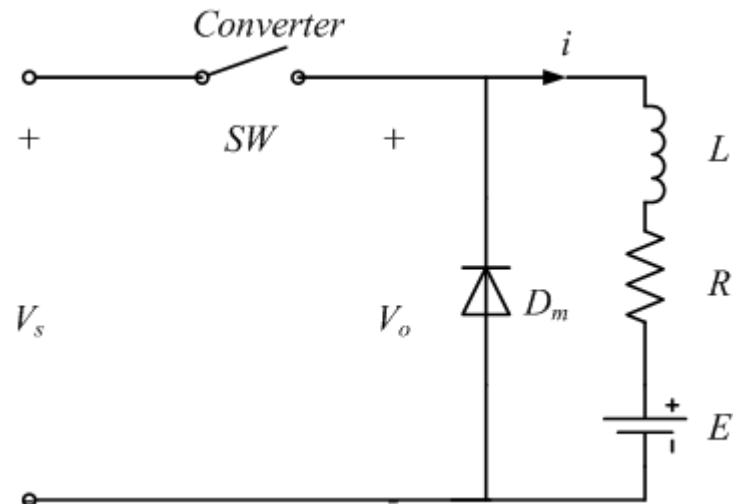
Step-Down chopper with RL Load:

➤ The operation of the chopper can be divided into two modes:

- During mode 1, the chopper is switched on and the current flows from the supply to the load.



- During mode 2, the chopper is switched off and the load current continues to flow through freewheeling diode D_m .



Case 1: Continuous load current:

For Mode 1 (the on-mode):

- The load current can be found from:

$$V_s = Ri_1 + L \frac{di_1}{dt} + E$$

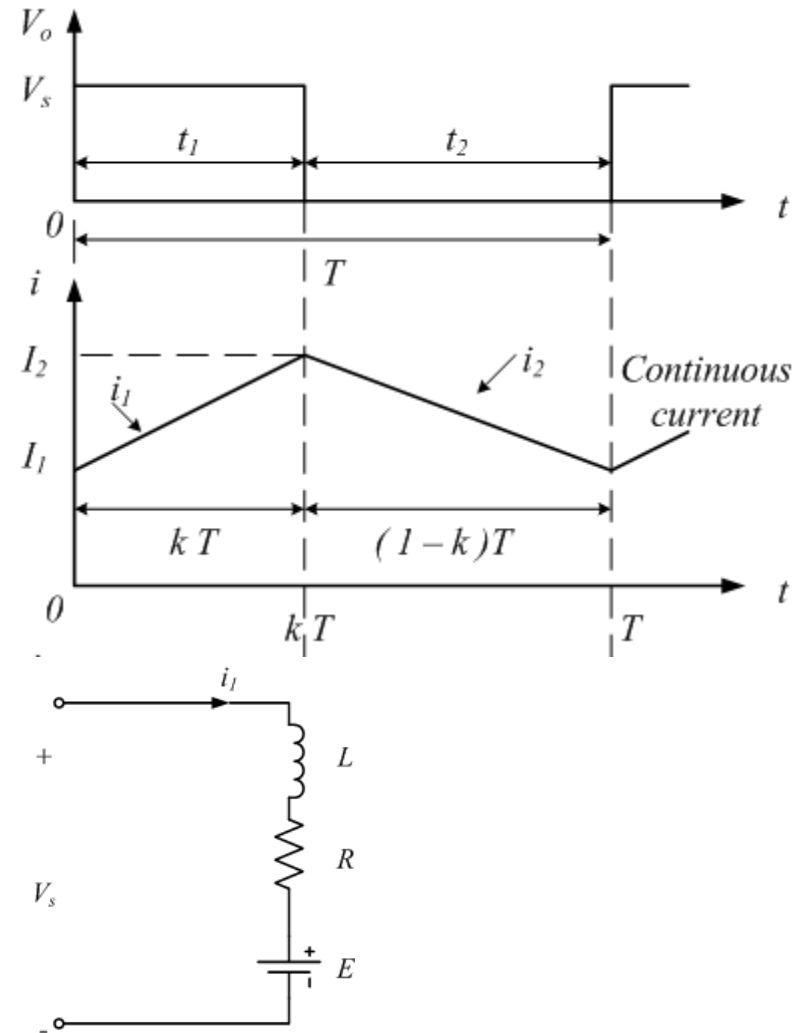
$$i_1(t) = Ae^{-\frac{tR}{L}} + \frac{V_s - E}{R}$$

- with initial current $i_1(t=0) = I_1$,
the load current is :

$$i_1(t) = I_1 e^{-tR/L} + \frac{V_s - E}{R} (1 - e^{-tR/L}) \dots (1)$$

- Mode 1 is valid during $0 \leq t \leq t_1 (=kT)$;
and at $t=t_1$, the load current becomes:

$$i_1(t = t_1 = kT) = I_2 \dots (2)$$



For Mode 2 (the off-mode):

- The load current in mode 2 is found from:

$$0 = Ri_2 + L \frac{di_2}{dt} + E$$

- with initial current $i_2(t=0) = I_2$, the load is:

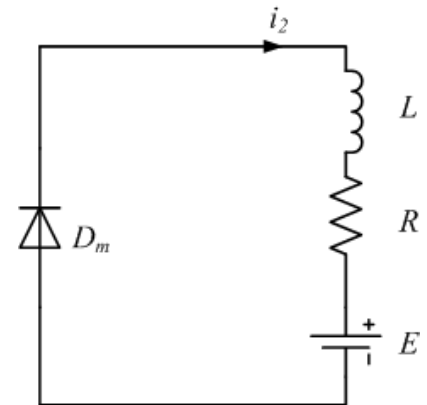
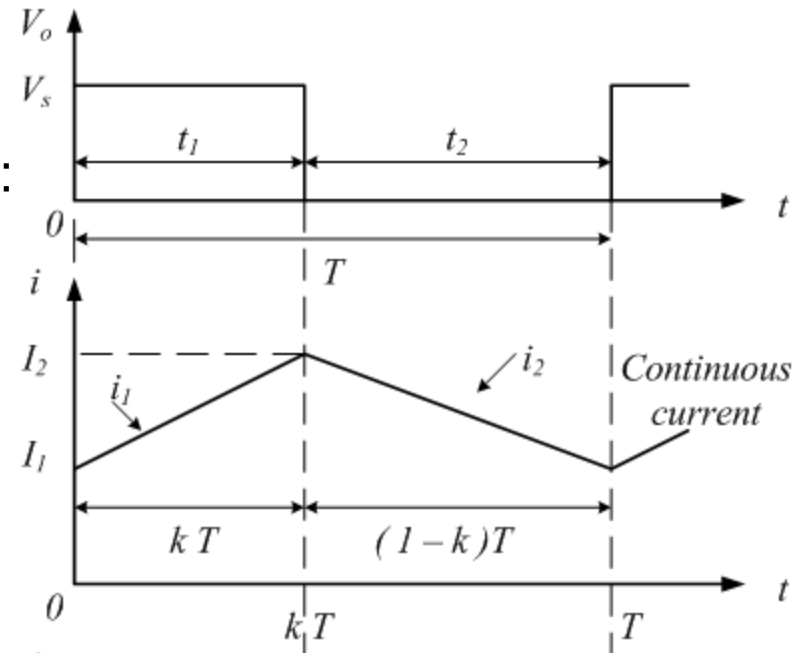
$$i_2(t) = I_2 e^{-tR/L} - \frac{E}{R} (1 - e^{-tR/L}) \dots\dots(3)$$

- Mode 2 is valid $0 \leq t \leq t_2 [= (1-k)T]$.

At the end of mode 2, the load current is:

$$i_2(t = t_2) = I_1 \dots\dots\dots(4)$$

- At the end of mode 2, the converter is turned-on again in the next cycle.



➤ The **peak-to-peak load ripple** current can be determined from Eqs. 1 to 4. Form Eqs. 1 and 2, I_2 is given by:

$$I_2 = I_1 e^{-kTR/L} + \frac{V_s - E}{R} (1 - e^{-kTR/L}) \quad \dots\dots\dots 5$$

➤ From Eqs. 3 and 4, I_1 is given by:

$$I_1 = I_2 e^{-(1-k)TR/L} - \frac{E}{R} (1 - e^{-(1-k)TR/L}) \quad \dots\dots\dots 6$$

➤ Solving for I_1 and I_2 , we get:

$$I_1 = \frac{V_s}{R} \left(\frac{e^{kz} - 1}{e^z - 1} \right) - \frac{E}{R} \quad \dots\dots\dots 7$$

$$I_2 = \frac{V_s}{R} \left(\frac{e^{-kz} - 1}{e^{-z} - 1} \right) - \frac{E}{R} \quad \dots\dots\dots 8$$

➤ Where $z = TR / L$ is the ratio of chopping or switching period to the load time constant

➤ The peak-to-peak current is $\Delta I = I_2 - I_1$

- Which after simplification becomes

$$\Delta I = \frac{V_s}{R} \frac{1 - e^{-kz} + e^{-z} - e^{-(1-k)z}}{1 - e^{-z}} \dots\dots\dots 9$$

- The condition for maximum ripple,

$$\frac{d(\Delta I)}{dk} = 0 \dots\dots\dots 10$$

- gives $e^{-kz} = e^{-(1-k)z} \Rightarrow -k = -(1-k) \Rightarrow k=0.5$. The maximum peak-to-peak ripple current (at $k=0.5$) is:

$$\Delta I_{\max} = \frac{V_s}{R} \tanh \frac{R}{4fL} \dots\dots\dots 11$$

- For $4fL \gg R$, $\tanh \theta \approx \theta$ and the maximum ripple current can be approximated to:

$$\boxed{\Delta I_{\max} = \frac{V_s}{4fL}} \dots\dots\dots 12$$

Note: Equations 1 to 12 are valid only for continuous current flow.

Case 2: Discontinuous load current:

➤ For low switching frequency ➔ large off-time ➔ the load current may decay to zero and becomes discontinuous

➤ In the on Mode, $I_1=0$ and Eq. 1 become:

$$i_1(t) = \frac{V_s - E}{R} (1 - e^{-tR/L}) \quad \dots\dots\dots (13)$$

➤ In the off-mode, Eq. 3 is valid for $0 \leq t \leq t_3$ such that $i_2(t = t_3) = I_1 = 0$, which gives:

$$0 = I_2 e^{-t_3 R/L} - \frac{E}{R} (1 - e^{-t_3 R/L}) \Rightarrow t_3 = \frac{L}{R} \ln \left(1 + \frac{R I_2}{E} \right)$$

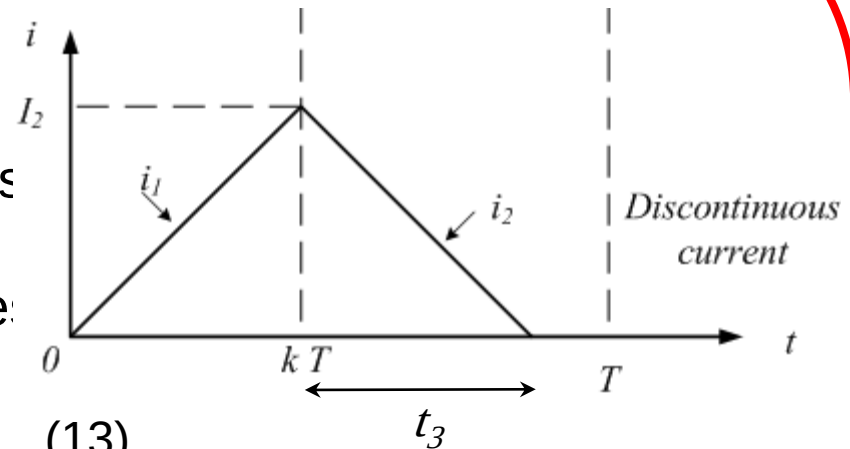
$$\therefore i_1(kT) = I_2$$

Substituting in Eq. 13

$$i_1(kt) = I_2 = \frac{V_s - E}{R} (1 - e^{-kz})$$

Substituting for I_2 in t_3 :

$$t_3 = \frac{L}{R} \ln \left[1 + \left(\frac{V_s - E}{E} \right) (1 - e^{-kz}) \right]$$



Case 2: Discontinuous load current:

- The average output voltage is calculated from:

$$V_a = \frac{1}{T} [kTV_s + (T - kT - t_3)E]$$

- The average load current is calculated as follows:

$$V_a = V_{L_average} + RI_a + E$$

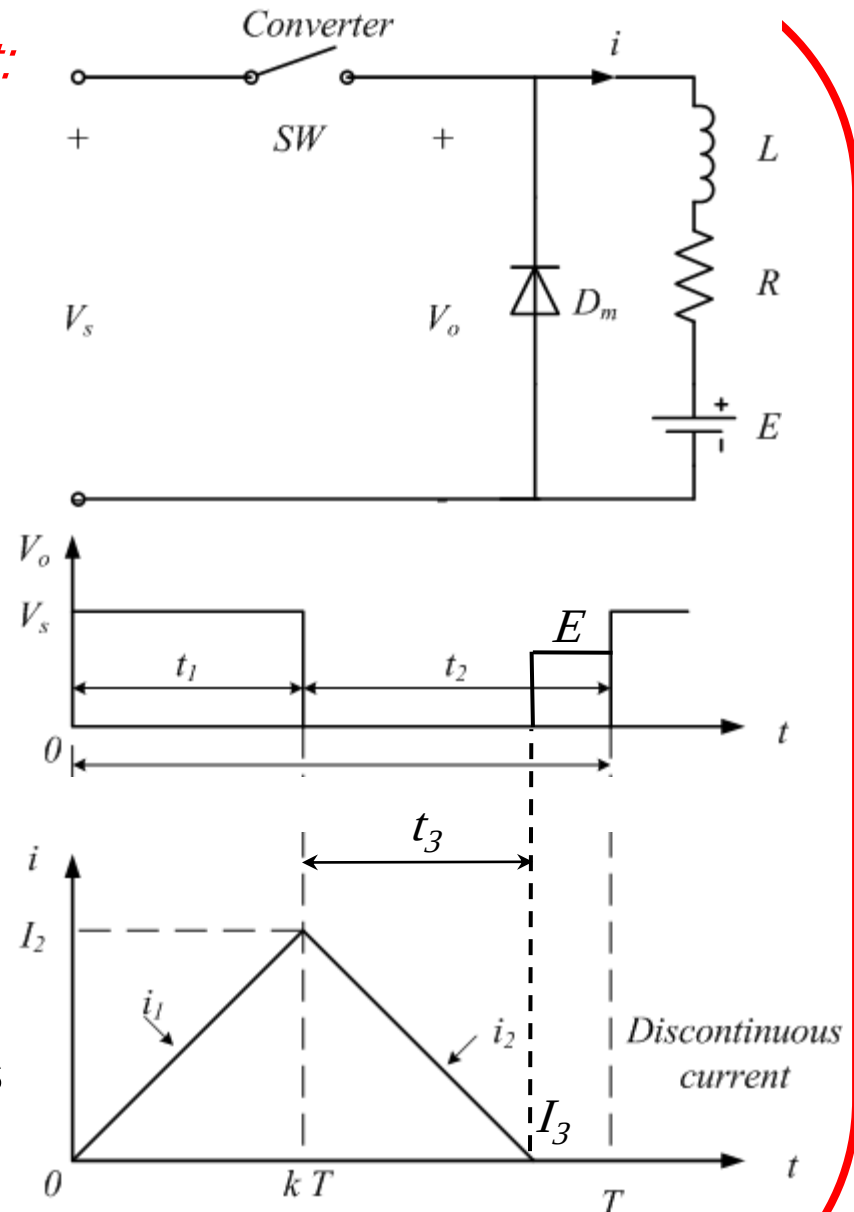
$$\therefore V_{L_average} = 0$$

$$\therefore I_a = \frac{V_a - E}{R}$$

This rule can be used for continuous current.

- The average supply current for lossless converter:

$$V_s I_s = V_a I_a \Rightarrow I_s = \frac{V_a I_a}{V_s}$$



Discontinuous current operation happen when:

➤ $E \uparrow$

➤ $\tau (=L/R) \ll T$

✓ $R \uparrow$

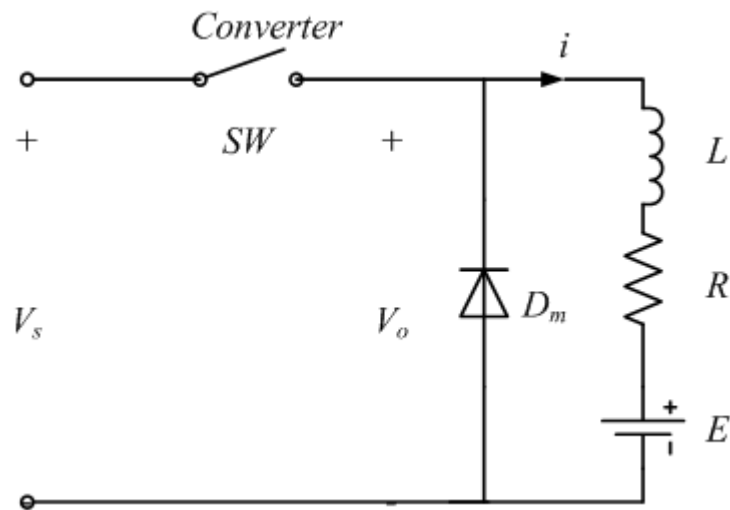
✓ $L \downarrow$

✓ $T \uparrow (f \downarrow)$

Example 2:

A converter is feeding an RL load as shown in the figure below with $V_s=220V$, $R=5\Omega$, $L=7.5mH$, $f=1kHz$, $k=0.5$, and $E=0V$. Calculate:

- (a) the minimum instantaneous load current I_1 ,
- (b) the peak instantaneous load current I_2 ,
- (c) the peak to peak load ripple current,
- (d) the average value of the load current I_a ,
- (e) the rms load current I_o ,
- (f) the effective input resistance R_i seen by the source,
- (g) the rms chopper current I_R .



Solution:

$$V_s = 220V, R = 5\ \Omega, L = 7.5mH, E = 0V, k = 0.5, f = 1kHz.$$

(a) The minimum instantaneous load current

$$I_1 = \frac{V_s}{R} \left(\frac{e^{kz} - 1}{e^z - 1} \right) - \frac{E}{R}$$

$$z = \frac{TR}{L} = \frac{0.001 \times 5}{7.5 \times 10^{-3}} = 0.6667$$

$$I_1 = \frac{220}{5} \left(\frac{e^{0.5 \times 0.6667} - 1}{e^{0.6667} - 1} \right) - \frac{0}{5} = 18.37A$$

(b) The maximum instantaneous load current

$$I_2 = \frac{V_s}{R} \left(\frac{e^{-kz} - 1}{e^{-z} - 1} \right) - \frac{E}{R}$$

$$I_2 = \frac{220}{5} \left(\frac{e^{-0.5 \times 0.6667} - 1}{e^{-0.6667} - 1} \right) - \frac{0}{5} = 25.63$$

Or:

By solving Equation (5) and (6) results in :

$$I_1 = 18.37\ A, \text{ and}$$

$$I_2 = 25.63\ A$$

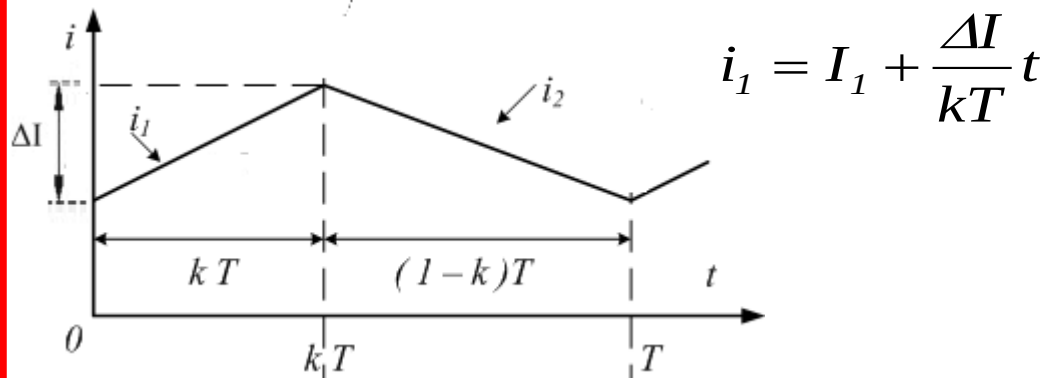
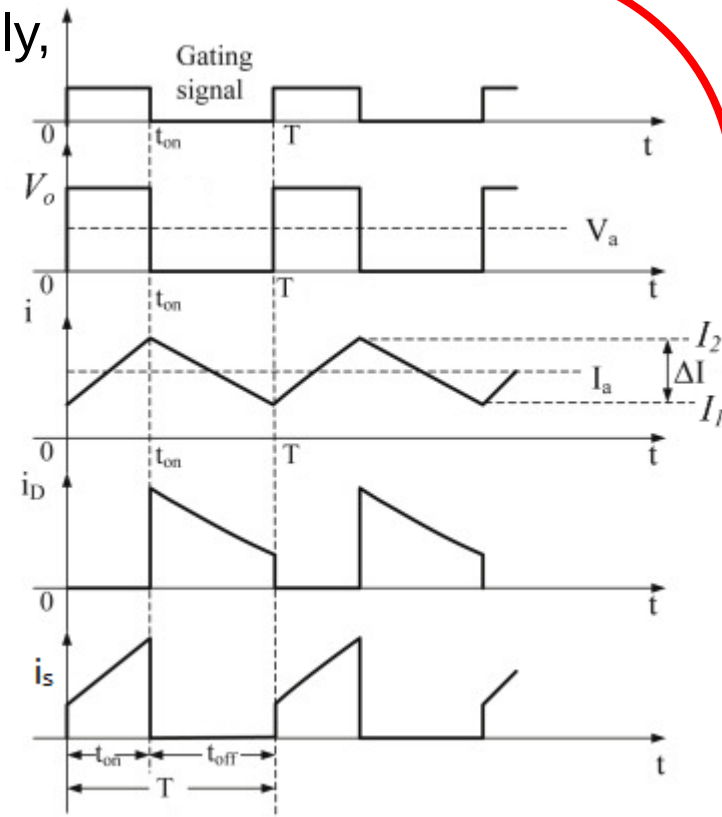
Proof is required.

(c) $\Delta I = I_2 - I_1 = 25.63 - 18.37 = 7.26A$

(d) The average load current is approximately,

$$I_a = \frac{I_1 + I_2}{2} = \frac{25.63 + 18.37}{2} = 22A$$

$$\text{or: } I_a = \frac{V_a}{R} = \frac{kV_s}{R} = \frac{0.5 \times 220}{5} = 22A$$



The rms value of the load current can be found from:

$$I_o = \sqrt{\frac{1}{kT} \int_0^{kT} i_1^2 dt} = \sqrt{\frac{1}{kT} \int_0^{kT} \left(I_1 + \frac{(I_2 - I_1)}{kT} t \right)^2 dt} = 22.1A$$

(f) The average source current:

$$I_s = k I_a = 0.5 \times 22 = 11A$$

And the effective input resistance,

$$R_i = \frac{V_s}{I_s} = \frac{220}{11} = 20\Omega$$

(g) The rms chopper current can be found from,

$$I_R = \sqrt{\frac{1}{T} \int_0^{kT} i_1^2 dt} = \sqrt{\frac{1}{T} \int_0^{kT} \left(I_1 + \frac{(I_2 - I_1)}{kT} t \right)^2 dt} = 15.63A$$